THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The difference between the squares of the 20th and 19th numbers is: 1

(a) 39 **(b)** 41 **(c)** 38 **(d)** 40

2. The smaller angle between the clock hands at 1 o'clock is 30°. The smaller angle between the hands at 11 o'clock is: 1

(a) 60 **(b)** 0 **(c)** 30 **(d)** 120

3. The sum of 10 consecutive counting numbers starting from 12 is: 1

(a) 166 **(b)** 165 **(c)** 164 **(d)** 175

4. The mean of the data 12, 18, 24, 30 is: 1

(a) 23 **(b)** 21 **(c)** 22 **(d)** 20

5. The greatest number that divides 48, 72 and 120 leaving no remainder is: 1

(a) 25 **(b)** 24 **(c)** 23 **(d)** 26

6. The first term of a pattern is 7 and the 14th term is 150. The common difference is: 1

(a) 11 **(b)** 10 **(c)** 13 **(d)** 12

7. The hour hand of a clock turns through how many degrees in 14 minutes? 1

(a) 8 **(b)** 12 **(c)** 6 **(d)** 7

8. A 5-digit number ends with digit 7 and has digit sum 31; the sum of its other four digits is: 1

(a) 23 **(b)** 24 **(c)** 33 **(d)** 25

9. Out of 120 students, 45 play cricket and 38 play football, none plays both. How many play neither? 1

(a) 37 **(b)** 42 **(c)** 38 **(d)** 36

10. For the 3-digit number 1d3, the digit d that makes it divisible by 9 (smallest such d) is: 1

(a) 14 **(b)** 5 **(c)** 7 **(d)** 6

11. A pattern adds 1, then 2, then 3, ... Starting at 10, the term after 12 additions is: 1

(a) 88 **(b)** 89 **(c)** 87 **(d)** 100

12. 4 lines are drawn through a point O. The number of straight angles at O (pairs of opposite rays) is: 1

(a) 3 **(b)** 4 **(c)** 12 **(d)** 5

13. The number of digits in the largest palindrome smaller than 10000: 1

(a) 5 **(b)** 3 **(c)** 6 **(d)** 4

14. A bar graph shows sales of ₹2000, ₹3500, ₹2500. If the scale is 1 cm = ₹500, the total height of all three bars is: 1

(a) 16 **(b)** 21 **(c)** 15 **(d)** 17

15. Which is the greatest 3-digit number that leaves remainder 3 when divided by 7? 1

(a) 990 **(b)** 998 **(c)** 1004 **(d)** 997

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. The sum of the first n odd numbers is 961. Find n and the 31th odd number. 2

17. A ray rotates 2 full turns and 90° more. Find the total angle turned and the number of right angles in it (nearest whole). 2

18. Find the number of terms in the Collatz sequence of 11 and its largest term. 2

19. A bar graph uses 1 cm = 20 units and a bar is 9 cm. Another bar is 3 cm shorter. Find the value of the shorter bar. 2

20. Show that the product of 3 consecutive numbers starting from 14 is divisible by 3. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Using the supercell idea, a row of 8 numbers starting 2, 3, then each next cell is the sum of the previous two. Find the 8th number and the sum of all 8 numbers. 3

22. In a pictograph one symbol = 12. A shop sold 3 symbols of pens, and the number of notebooks is 2 times the pens. Find the notebooks sold and total items sold. 3

23. Find the greatest number that divides 76 and 100 leaving remainder 4 in each case. 3

24. A tower is built with rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 6, and in the first 6 rows together? 3

OR

A pizza is cut by 2 lines through the centre into equal slices (all lines passing through the centre). Find the angle of each slice and the number of slices. 3

25. How many 4-digit numbers can be formed from 2, 5, 7, 9 using each digit once? What is the largest and the smallest? What is their difference? 3

OR

For the data 5, 5, 2, 7, 8, 3, 5, 1, 7, 4, 6, 5 prepare a frequency table and find how many values are above the mean and the mean itself. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A rectangular garden is 60 m by 36 m. Square flower beds of the largest possible equal size are to be made covering the garden. Find 5

(i) the side of each bed,

(ii) the number of beds,

(iii) the cost of fencing all beds at ₹5 per metre if each bed is fenced separately.

OR

Two sequences: A is $5, 9, 13, 17, \ldots$ and B is $20, 22, 24, 26, \ldots$. Find the 12th term of each and decide which is larger and by how much. **5**

27. A robot starts facing North. It turns clockwise $180°$, then $45°$ anticlockwise, then $90°$ clockwise. Find its final direction angle clockwise from North and the further turn needed to face North again. **5**

OR

Find the number of steps the reverse-and-add process takes from 67 to reach a palindrome, write each sum, and state whether the final palindrome has an even or odd number of digits. **5**

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND